

1. Electrochemical model

SPMe (PyBaMM)
Eqs (1)–(4): particle diffusion,
Butler–Volmer
→ voltage $V(t)$

2. Degradation submodels

SEI, plating, LAM, cracking
coupled to the model
(full equations in the SI)

3. Observables

capacity $Q(n)$,
low-rate dV/dQ ,
EIS $Z(\omega)$

Ground truth (hidden)

per-mechanism loss
 $Q_m(n)$, Eq. (5)
(used only to score)

4. Identification

features → k -NN
classify mechanism /
regress fractions

5. Results

confusion matrix, R^2 ,
identifiability map

score (never an input)